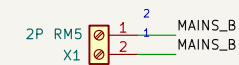
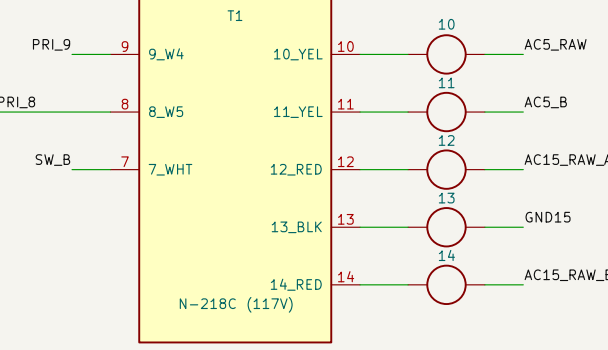
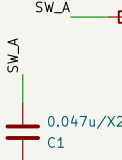
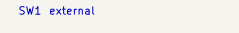


MAINS INPUT, SWITCH, TRANSFORMER

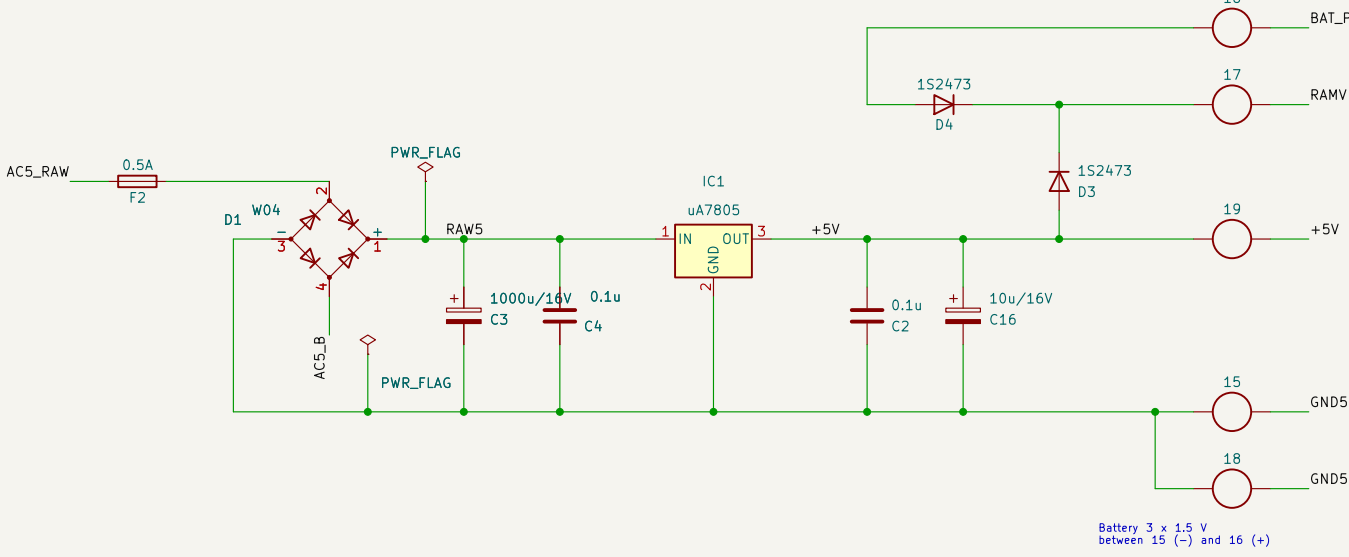
Terminals 1-9: RMS screw terminals (110 V primary side)



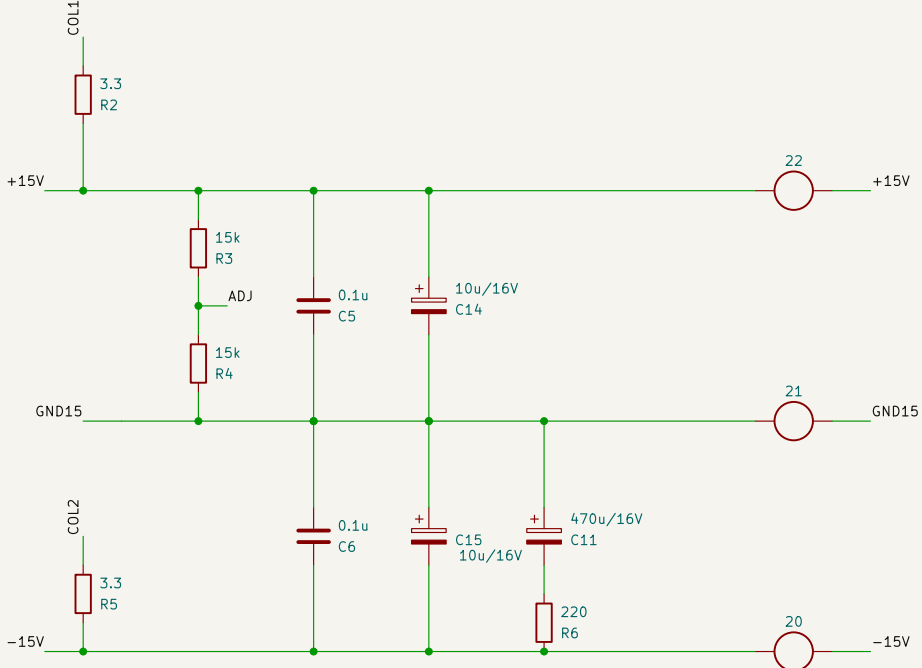
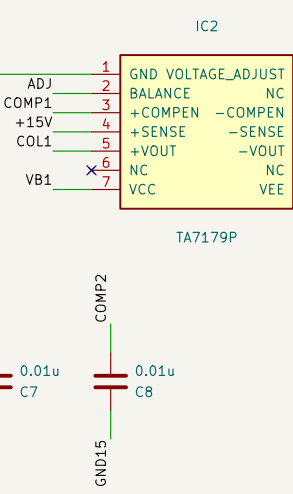
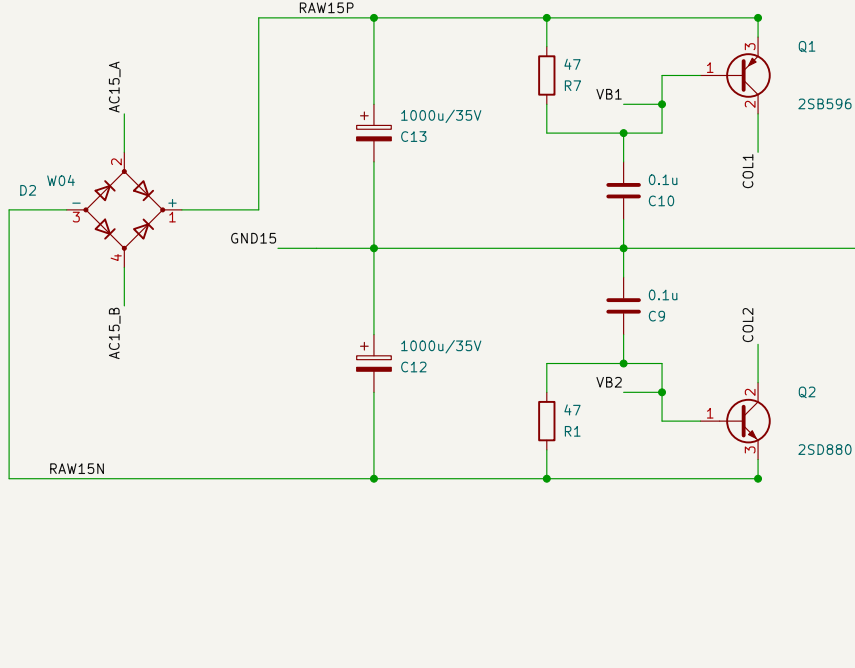
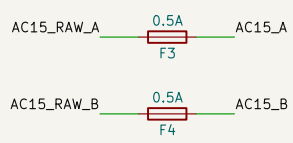
Mains (W1) on 3/4, mains (W3) on 1/2;
mains switch SW1 between 1-5 and 2-6;
transformer T1 on 7-9 and 10-14



+5 V BRANCH (IC1, RAM BACKUP)



+ -15 V BRANCH (IC2 TA7179P, Q1, Q2)



MOUNTING: 4 board holes, 6 heat sink holes (246-101A), 3 screw holes for the regulators



UNTESTED - NOT MANUFACTURED, NOT MEASURED, NOT SAFETY-TESTED. MAINS VOLTAGE (110 V) CAN KILL.

ROLAND TR-808 - POWER SUPPLY PS3116-051 (100/117 V, built for 110 V), PCB 291-405A

Redrawn from the service manual (15 June 1981); schematic sheet 13, component layout sheet 12, parts list sheet 10. Designators and terminal numbers 1-22 as in the original.

Bridge rectifiers D1/D2, W04 (400 V, 2W04G on the unit); the manual gives W-02 (200 V); the W04 fits the same package and is a permissible replacement.

110 V variant (PS3116-051, 117 V tap): C1 ECQ-UC1A473MC 0.047 uF, fuses 5GA 0.5 A, transformer N-218C (117 V), mains switch SDG5P-001. 110 V is 6 % below the 117 V rating. The 220/240 V variant (PS3116-054) uses the same board.

Secondary voltages at 0.15 A DC: terminals 10-11 (YEL-YEL) 10 V DC, 12-14 (RED-BLK) 23 V DC (per half). Outputs: +5 V (19), +15 V (22), -15 V (20), 0 V (21), RAM supply +5 V / battery 3 x 1.5 V (16, 17).

The two grounds (5 V branch: 15/18, +-15 V branch: 13/21) are NOT connected on the board in the original; they meet only outside the board. Transformer T1, mains switch and battery are not on the board (terminals 7-14, 4-6, 15/16). The wire jumper J1 of the original (single-sided board) is dropped; this rebuild is two-layer. Terminals 1-9 (110 V primary side) are RMS screw terminals X1-X4; 10-22 are solder lugs.

Redrawn from the Roland TR-808 service notes, 15 June 1981

Sheet: /
File: TR-808PSU_Rev0.4-110V.kicad_sch

Title: TR-808 PSU PS3116 rebuild - UNTESTED

Size: A2 Date: 2026-10-10 Rev: 0.4

KiCad E.D.A. 10.0.2 Id: 1/1